

D1: Sensor Count and Layout

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What does each additional sensor buy, and where should it go?

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SHARC buoy project Code: `runD1SensorLayout.m`, commit `e9eacc7` Results: `Results/D1/D1_20261008_175233`

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 2 Oct run were low by $\alpha(\omega) = H\omega^2/g$; `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08). Settings unchanged. This summary supersedes D1_20261002_145943. Because α grows with frequency, the correction reweights the band towards high frequencies, so the optimal layouts can change as well as the numbers. They did, from $n_s = 3$ upward.

Summary

Question. For $n_s = 2, 3, 4$ sensors on the 33-point candidate grid, which layout and reference minimise the parameter uncertainty of the inverse, how much does each extra sensor buy, and how do the Level 2C and Gramian designs compare? **Outcome (LSM6DSV16X, p_0 , single known wave direction).** **Two sensors suffice:** the best pair (s6 and s18, on the incidence axis) gives $\sigma_{\ln \beta} = 0.063\%$ and is Identifiable, but with $d_W = 0.476$ it is close to the bias limit. A third sensor lowers $\sigma_{\ln \beta}$ by 16.0% and a fourth by 19.6%, both below the pre-registered 20% rule. The optimum now leaves the axis at $n_s = 3$ (s28, edge ring, 90°) and is no longer nested at $n_s = 4$. The reference choice still does not matter (median $\times 1.00$). Level 2C remains a good design (rank 933 of 40 920, $\eta = 0.72$); the Gramian $\lambda_{\min}$ designs are poor ($\eta = 0.07$ to 0.24). **Extra sensors buy robustness**, both to placement and to the wave heading; D3 quantifies the latter.

1 What the study does

Fisher information per layout. For a layout $\mathcal{L}$ with reference r , the features are the real and imaginary parts of $T_j = S_{jr}/(S_{rr} - S_{nn})$ for the $n_s - 1$ non-reference sensors, and

$$F(\mathcal{L}, r) = \sum_k J_k^T \Sigma_k^{-1} J_k, \quad \sigma_{\ln \beta}^2 = (F^{-1})_{11}, \quad \sigma_{\ln R}^2 = (F^{-1})_{22},$$

with Σ_k the full shared-reference covariance (complex Wishart, $N_{\text{eff}} - 1$ degrees of freedom) of the E1 engine. The field H and $\partial H / \partial \ln p$ are computed once at all 33 points (loaded from the migrated cache); each layout extracts its rows, so the 181 104 (layout, reference) evaluations per noise case need no EMM solves.

Criteria. Primary: A-optimal, minimise $A = \text{tr}(F^{-1}) = \sigma_{\ln \beta}^2 + \sigma_{\ln R}^2$. Secondary: D-optimal ($\log \det F$) and E-optimal ($\lambda_{\min}(F)$). Each criterion chooses its own best reference. Efficiency of a design: $\eta = A_{\text{opt}}/A_{\text{design}} \in (0, 1]$. Worth-it rule (pre-registered): sensor $n + 1$ is worth adding if $G = 1 - \sigma_{\ln \beta}(n + 1)/\sigma_{\ln \beta}(n) \geq 20\%$ at the optimum. Recommended count: the smallest n_s whose optimum is Identifiable at LSM6DSV16X noise.

Setting. p_0 ($R_{\text{phys}} = 0.72$ m), setting F ($N = 65536$, $L = 2048$, 84 bins in 3.178 to 8.301 rad/s), JONSWAP $H_s = 0.05$ m, single wave direction along the incidence axis. Grid: centre s1, then rings at $r/R = 0.3, 0.5, 0.7, 0.9$ (s2 to s9, s10 to s17, s18 to s25, s26 to s33) at eight angles $0, \pi/4, \dots, 7\pi/4$. Admissibility: $r \leq 0.9R$ and spacing ≥ 0.10 m; the closest grid points are 0.144 m apart, so every layout is admissible (528, 5456 and 40 920). Spacing sensitivity: 0.05 and 0.15 m. Noise: LSM6DSV16X (headline), ICM-42688-P and G3.

Checks. Gate: Level 2C with s26 matches `evaluateScenario` in all three noise cases (relative differences $\leq 7 \times 10^{-12}$); all 84 bins accepted and all blocks positive definite for every reference. Mirror symmetry about the incidence axis holds to 1.6×10^{-11} . The Gramian optima are recomputed exactly as in `level2C finalCheck` ($N = 4001$); they do not depend on the FRF amplitude.

2 Results

Optimal designs (LSM6DSV16X). Gain G is relative to the $n_s - 1$ optimum.

n_s	A-optimal layout	Ref	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$	d_W	$\kappa(F)$	G	Class
2	s6, s18	s6	0.063%	0.0145%	0.476	36	–	Identifiable
3	s6, s18, s28	s6	0.053%	0.0104%	0.327	56	16.0%	Identifiable
4	s6, s14, s26, s28	s6	0.042%	0.0075%	0.467	69	19.6%	Identifiable

n_s	D-optimal	$\sigma_{\ln \beta}$	E-optimal	$\sigma_{\ln \beta}$	Spacing 0.15 m (A)	$\sigma_{\ln \beta}$
2	s6, s26	0.066%	s6, s18	0.063%	unchanged	0.063%
3	s6, s26, s30	0.060%	s6, s18, s28	0.053%	unchanged	0.053%
4	s6, s7, s26, s30	0.050%	s6, s14, s26, s28	0.042%	s6, s15, s26, s28	0.043%

Spacing 0.05 m and unconstrained: identical to the 0.10 m optimum. Other noise cases, A-optimum: ICM-42688-P s6 s18 / s6 s14 s26 / s6 s14 s26 s28 with $\sigma_{\ln \beta} = 0.079, 0.067, 0.056\%$; G3 the same layouts with 0.077, 0.065, 0.055%; all Identifiable, recommended $n_s = 2$.

Pre-amendment (2 Oct), LSM6DSV16X: $\sigma_{\ln \beta} = 0.463, 0.372, 0.334\%$, optima s6 s18 / s6 s14 s18 / s6 s14 s18 s26.

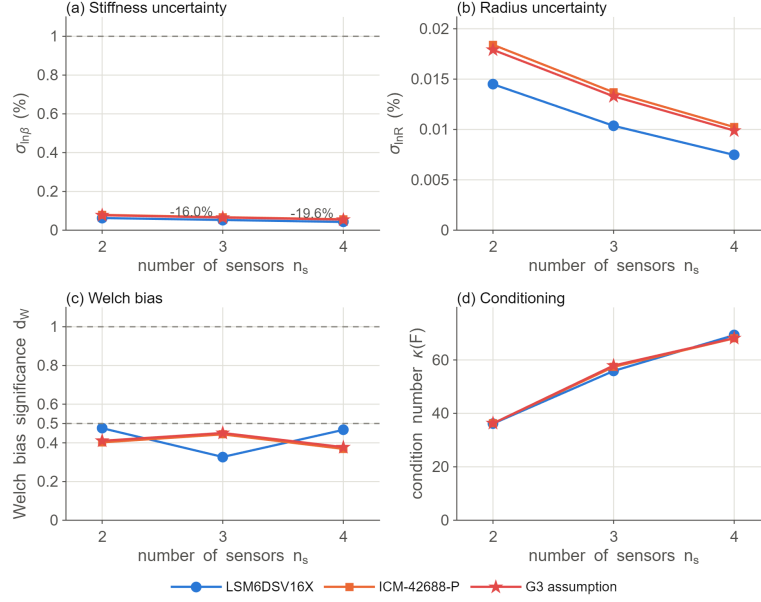


Figure 1. Best achievable (a) $\sigma_{\ln \beta}$, (b) $\sigma_{\ln R}$, (c) Welch bias significance d_W and (d) condition number $\kappa(F)$ against n_s , at the A-optimal layout, for three noise levels. Annotations in (a): gain per added sensor (LSM6DSV16X). Dashed lines: the 1% and $d = 0.5$, 1 class thresholds.

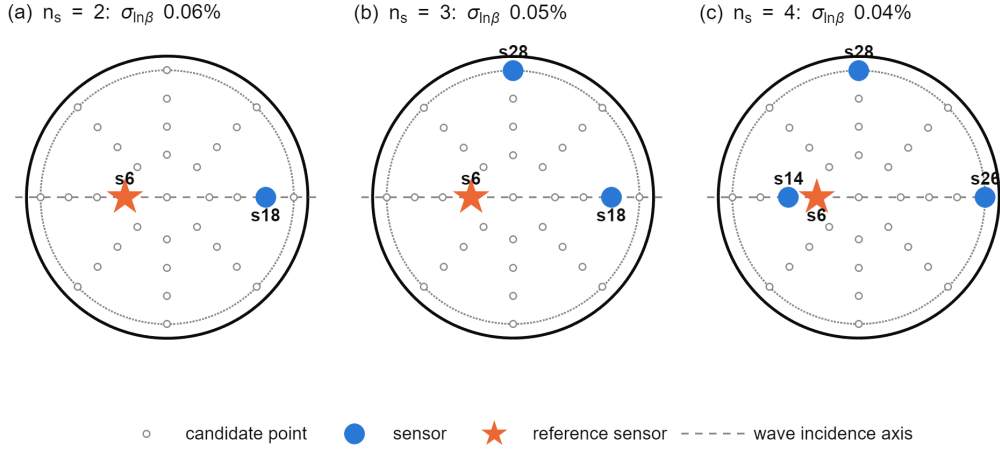


Figure 2. A-optimal layouts for $n_s = 2, 3, 4$ (LSM6DSV16X), viewed from above. Open circles: candidate points; dotted circle: $r = 0.9R$ admissibility limit; dashed line: wave incidence axis.

How forgiving is placement? Distribution of $\sigma_{\ln \beta}$ over all layouts, each with its best reference.

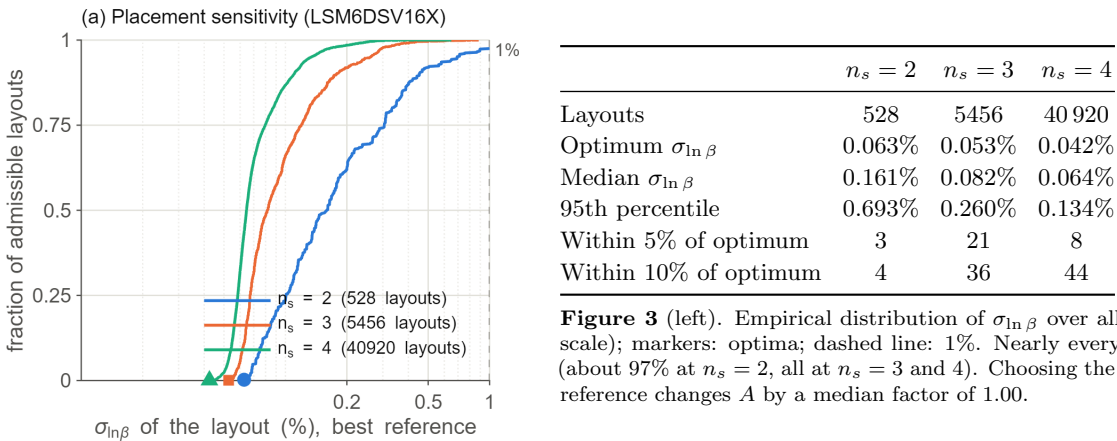


Figure 3 (left). Empirical distribution of $\sigma_{\ln \beta}$ over all admissible layouts (log scale); markers: optima; dashed line: 1%. Nearly every layout now reaches 1% (about 97% at $n_s = 2$, all at $n_s = 3$ and 4). Choosing the best instead of the worst reference changes A by a median factor of 1.00.

Comparison with earlier designs ($\eta = A_{\text{opt}}/A_{\text{design}}$; rank in the A ordering, best reference).

Design	Layout	Rank	η
Level 2C ($n_s = 4$)	s2, s6, s14, s26 (best reference s6; s26: rank 934)	933 / 40 920	0.722
Gramian $\lambda_{\min}$, $n_s = 2$	s26, s30	362 / 528	0.071
Gramian $\lambda_{\min}$, $n_s = 3$	s26, s27, s30 (2 tied)	4565–4566 / 5456	0.132
Gramian $\lambda_{\min}$, $n_s = 4$	s26, s27, s29, s30 (4 tied)	32 508–32 511 / 40 920	0.243

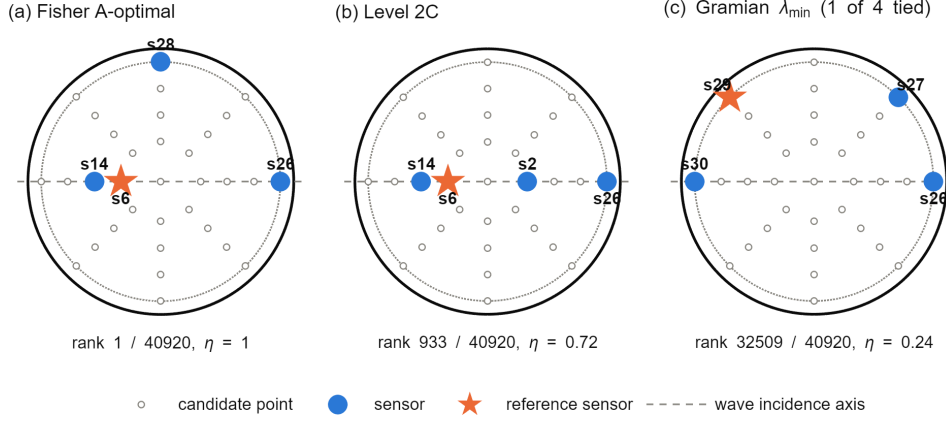


Figure 4. Four-sensor designs compared: (a) Fisher A-optimum, (b) Level 2C, (c) one of the four tied Gramian $\lambda_{\min}$ optima, with their rank and efficiency η in the Fisher A ordering (LSM6DSV16X).

Heading sensitivity of the optima (efficiency A_{opt}/A of each optimum rotated about the centre, known heading; printed by the runner):

Rotation	0°	45°	90°	135°	180°	225°	270°	315°
$n_s = 2$: s6, s18	1.00	0.71	0.07	0.26	0.28	0.26	0.07	0.71
$n_s = 3$: s6, s18, s28	1.00	0.68	0.31	0.43	0.36	0.59	0.24	0.59
$n_s = 4$: s6, s14, s26, s28	1.00	0.51	0.34	0.35	0.27	0.36	0.24	0.48

3 Interpretation

1. **Two sensors are enough under the pre-registered rule, but the class is now set by bias.** The optimal pair has $\sigma_{\ln \beta} = 0.063\%$, about sixteen times inside the 1% limit, so precision is not the constraint; $d_W = 0.476$ is, with little margin to 0.5 (as in D2). The third and fourth sensors gain 16.0% and 19.6%, both below 20%, at all three noise levels (ICM-42688-P and G3: about 15% each).
2. **Geometry of the optimum has changed.** The pair is still on the incidence axis (s6 at $0.3R$, s18 at $0.7R$ on the opposite side). The third sensor is now **off-axis**: s28 on the edge ring at 90° . At $n_s = 4$ the optimum is s6, s14, s26 (on-axis) plus s28, which drops s18, so the designs are nested only up to $n_s = 3$. For ICM-42688-P and G3 the three-sensor optimum stays on-axis (s6, s14, s26) and is nested with the four-sensor one. A plausible reading: the α correction weights the high-frequency bins more, where shorter flexural wavelengths make cross-axis ($\cos n\theta$, $n \geq 1$) structure informative, and an edge sensor at 90° samples it. Several layouts are close (21 within 5% at $n_s = 3$), so the exact optimum is not sharply defined.
3. **Extra sensors buy robustness.** The optimum improves modestly with n_s , but the distribution tightens: the median falls from 0.161% to 0.064% and the 95th percentile from 0.69% to 0.13%. Rotated by 90° against the wave, the two-sensor optimum keeps only 7% of its efficiency; three and four sensors keep at least 24% at any heading. A deployment guarantees neither the heading nor exact positions.
4. **The reference choice does not matter for variance.** With the full shared-reference covariance, changing the reference re-expresses the same cross-spectral information (median ratio 1.00). The optima now select s6 instead of s18 or s26, which is immaterial; choose the reference on practical grounds (highest SNR, R3).
5. **Level 2C is still a good design.** Rank 933 of 40 920 (top 2.3%), $\eta = 0.72$, down from 0.89 because the optimum moved off-axis: Level 2C shares s6, s14 and s26 with the four-sensor optimum and has s2 instead of s28. Its production reference s26 is within one rank of the best reference (s6).
6. **Modal observability is not parameter identifiability.** The Gramian $\lambda_{\min}$ designs place sensors on the edge ring where the two modes are most observable and rank in the bottom third ($\eta = 0.07$ to 0.24). The Gramian is not a usable proxy for this inverse.
7. **Criterion and spacing.** E-optimal designs coincide with the A-optimal ones at every n_s ; D-optimal designs favour the edge and cost 5 to 19% in $\sigma_{\ln \beta}$. A 0.15 m spacing rule costs 2% at $n_s = 4$ and nothing below.
8. **Bias is not monotone in n_s .** d_W is 0.476, 0.327 and 0.467 for $n_s = 2, 3, 4$. A-optimality ignores the Welch bias, so the A-optimal layout need not have the most bias margin; the three-sensor optimum has the most room for the systematic errors of R3, R4, V1 and V2. A bias-aware selection (minimum A subject to $d_W \leq$ a margin) is a natural refinement and is how D3b already selects.

Limitations.

- **Single, known wave direction.** The heading table shows how strongly the optima depend on it; D3 repeats this search with the heading as a nuisance parameter and under directional seas (V2 model).
- **One floe, one sea state.** p_0 and JONSWAP $H_s = 0.05$ m; A1 and R2 test other floes and spectra.
- **Grid, positions, linearisation.** 33 candidate points; position errors in V1; M1 checks the two-sensor optimum.

Requirement for the experiment and the thesis

Under a known, single wave direction, **two sensors on the incidence axis** (s6 at $0.3R$, s18 at $0.7R$ on the opposite side) meet the Identifiable class with LSM6DSV16X, ICM-42688-P or G3 noise, with $\sigma_{\ln \beta} \approx 0.06$ to 0.08% but d_W close to 0.5 . Because the pair loses most of its information if the heading is wrong, deploy **three or four** sensors; the final layout is set by the D3 rerun. Level 2C remains acceptable ($\eta = 0.72$). Choose the reference by SNR, not by variance. Do not use the Gramian criterion to place sensors for this inverse.

Outputs (Results/D1/): D1_20261008_175233.log/.mat/_summary.txt; D1_optimum.csv, D1_sensor_count.csv, D1_distribution.csv, D1_comparison.csv; figures/D1_fig1 to D1_fig4 (.pdf/.png). Superseded: D1_20261002_145943 (pre-amendment).